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Prologue to the Aims and Achievements of Eleventh Indian Expedition to Antarctica

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**"Ah, but a man's reach
Should exceed his grasp
Or what's a heaven for"**

(R. Browning)

Introduction

Since 1981, when India focussed her attention on Antarctica, It has been a story of progress in leaps and bounds. Contributions made by Indian scientists within a short period of only a decade, in various disciplines of Antarctic science, are phenomenal. It was with an intention of continuing this process of providing valuable inputs to the existing knowledge about Antarctica, that the eleventh scientific expedition to this distant land was launched in the austral summer of 1991-92. The fury of nature gave rise to an unprecedented situation in the context of Indian expeditions, restraining the ship from going alongside ice shelf till as late as 22 February 1992. Yet, almost all the assigned tasks were completed. Some of the achievements are the geological investigation of Payer and Weyprecht mountains - hitherto a virgin area, recce for future camping sites in Conrad and Orvin mountains, GPS observations to set up control points in all the ranges of Wohlthat mountain chain from Gruber to Humboldt, systematic and detailed study of Antarctic and other birds, studies on Planetary Boundary Layer, detailed study on lichens of Schirmacher Oases, large scale topographical mapping of the area surrounding Maitri station, etc. Erection of a 28 m tower for scientific studies and communication purposes, strengthening of water supply line and pumphouse structure, installation of new generators, experiments conducted on HF communication as well as on amateur radio bands and record number of flying hours clocked by helicopters despite bad weather conditions, are highlights from logistic sector.

Tasks

The summer time scientific and logistic assignments for the team are outlined below:

Scientific tasks

1. Geological investigation of Payer and-Weyprecht mountains in Wohlthat range.
2. Routine glaciological observations including monitoring of the snout of Dakshin Gangotri Glacier in Schumacher Oasis.
3. Search for meteorites.
4. Large scale topographical mapping of Maitri station area.
5. GPS observations in Schinnacher Oasis and Wohlthat mountains for ground control points to prepare digital terrain model with the help of SPOT imageries.
6. Measurement of aerosol concentration over Antarctica.
7. Collection of data to study the Planetary Boundary Layer (PBL).
8. Investigation in marine and fresh water eco-systems in the Antarctic environment.
9. Studies on lichenometry in Schirmacher Oasis, Antarctica.
10. Study of bacteria in Antarctic soil collected from Schirmacher Oasis.
11. Investigation of faunal aspects in Antarctica.
12. Studies on circadian rhythm changes on human beings in Antarctic environment.
13. Routine studies in geomagnetism.
14. Routine meteorological studies.
15. Studies on integrated water vapour content of atmosphere and radio noise measurements.

Logistic tasks

16. Enhancing fire protection measures in Maitri station.
17. Installation of new generator sets in the genset accommodation and commissioning of power house.
18. Improving the heating system in Maitri station.
19. Improving the structure for water supply system at Maitri station.
20. Erection of 28 m tower for fixing scientific equipments and communication antennas.
21. Routine maintenance of vehicles.
22. Erection of living and toilet modules in summer camp.
23. Establishing data transfer facility using HF link with India.

The Voyage

Expedition ship M.V.THULELAND sailed at 1800 hrs from Mormgao harbour, Goa on 27th of November 1991 with 98 members on board. Of these, 29 were scientists Including 7 member who were to winterover the Indian station Maitri, drawn from various government organisations, universities and national laboratories. Out of the rest, 65 members formed the logistic support team with 28 personnel drawn from Indian Army, 18 from Indian Navy and 19 from Indian Air Force. There was one representative each from Department of Ocean Development, Films Division and field of journalism. The team had also a meteorologist member from Colombia as guest scientist who was sent by his country basically to prepare for a future Colombian expedition to Antarctica.

The journey was fairly smooth upto the outskirts of Antarctica. Enroute, the equator was crossed on 30 November 1991. There was a halt at Port Louis, Mauritius from 05 December to 07 December 1991 , where a helicopter belonging to Mauritian Police, which had undergone repairs in India, courtesy Indian Government, was delivered and fresh fruits and vegetables were purchased. The team also received some packages of spares which could not be supplied at Goa.

All through the onward journey scientific observations went on uninterrupted which comprised meteorological observations, marine biological studies, bird logging, medical research and iceberg monitoring (in cold waters). The EME team utilised the journey time by repairing the two SNOCAT and one PLSTEN BULLY vehicles which were taken on board in unserviceable conditions. While scientists delivered popular talks in their respective fields for the benefit and knowledge of others, the logistic personnel apprised their fellow expeditioners with the working of vehicles, helicopters, generators and other equipments being used in Antarctica. This fruitfully contributed in maintaining a very high and healthy morale of team members for the entire duration of nearly one month's stay in sea.

The Landing

**“The ice was here, the ice was there
The ice was all around
It cracked and growled
And roared and howled
Like noises in a swound”**

(S.T.Coleridge) in “Rhymes of Ancient Mariner”

macher Oasis. Mapping of the area around Maitri station was also earned out on 1:5,000 scale with a contour interval of 5 metres.

IIG continuously monitored changes in the geomagnetic field along three orthogonal components X, Y and Z with the help of a fluxgate magnetometer. For a short duration, three stations were operated simultaneously at Maitri, Dakshin Gangotri and Payer, located at the vertices of a triangle of 100 to 250 km a side. Later on, however, Payer camp located In the mountains had to be closed due to bad weather while the other two stations continued functioning satisfactorily.

Bio-sciences

ZSI conducted bird logging, ethiological experiments, faunal survey and ringing of birds besides audio/video recording of faunal elements, and study of soil microbes. Altogether seven Antarctic skuas were ringed (ring numbers F 13100 to F 13106). In addition, eleven Adelie penguins were also ringed (ring numbers K 591 to K 601). A total of 21 blood samples were collected from seven skuas and 17 from 11 Adelie penguins for hematological and eitdoparasi-tic studies. The present volume includes a detailed report on the 46 species of oceanic elements identified during the expedition.

NBRI scientist collected about 150 samples of lichens aiongwith their substrate from Schirmacher Oasis. They have been studied morphologically and anatomically resulting in the identification of 13 genera and 21 species. In-addition, samples were collected to estimate percentage accumulation of radio nuclides and heavy metals in the lichens of this region.

Experiments conducted by BHU were directed towards determining nitrogen content of Antarctic soil and isolation and characterization of cyanobacteria present in soil and water samples of Schirmacher Oasis.

Scientist from DARL carried out studies on micro and macro fauna of Schirmacher Oasis, production of harvestable products for food in Antarctic environment under variable glass house conditions, and on use of cyanobacteria as biofertilizer in vegetable cultivation by hydroponic system in Antarctica.

Investigations were carried out by NIO on marine and fresh water eco-system in Antarctica which comrised particulate flux and carbon budget in the Antarctic coastal eco-system, growth stimulating factors for phytoplankton production, association of increased UV-radiations with growth depletion, nutrients and biological productivity at ice edge zone, coupling of microbial and bacterial biomass, oceanic front and biological productivity, etc.

We crossed 40° South latitude on 12 December 1991 and entered into the pack ice around 18 December. The helicopters were taken out from ship's hold, their rotors assembled, test flights completed and kept ready for immediate operation on arrival in polynya. The ship reached polynya on a fair weather evening of 23 December 1991 and almost immediately a flight was taken to Maitri station to the delight and cheers of the overwintering members led by Dr.A.K.Hanjura. However, weather deteriorated fast within hours of our landing at Maitri and thereafter, actual task of shifting scientists and logistic personnel to their respective working areas could begin only on 28 December 1991 frequently interrupted by spells of bad to very bad weather. A thick fast ice of more than 20 km width kept our ship at bay from reaching the edge of ice shelf till late February.

At. Antarctica

Before our ship could finally come alongside the ice shelf on 22 February 1992, off loading of bulk fuel and the three vehicles plus a dozer was not possible. However, rest of the tasks were carried out facing a constant challenge thrown by the weather gods. Following is a summary of the work done :-

Earth Sciences

GS1 set up a camp in Payer mountains at 71°51'26"S : 14°28'54"E at an altitude of 1995 m a.s.l. on 03 January 1992. Field work was done by foot traverses within approachable limits and by air support in the outer peripheries of the area. About 100 rock samples from different observation points were collected for detailed petrography and geochemical studies. Approx, 1100 sq km area was geologically mapped which includes exposures in northern Payer and Weyprecht mountains, nunataks around them and intervening ice. This is the first record of a geological camp being established in this area. Search for meteorite between southern limits of Humboldt mountains and 72°36'30" South latitude was also carried out.

The snout of Dakshin Gangotri Glacier was monitored with the help of SOI in addition to the studies on accumulation/ablation pattern of snow over ice shelf.

While at sea, GSI team had carried out systematic monitoring of the icebergs following the guidelines of Norwegian Polar Institute.

SOI set up seven GPS control points - two in Gruber massif, one each in Petermann I,II,III mountains, one in Humboldt mountains and one in Schir-

Physical Sciences

NPL in collaboration with ISI carried out studies on atmospheric and planetary boundary layers (ABL and PBL) over Antarctica and also measured actual concentration and radiation over Sdhurmacher Oasis for understanding the cause behind ozone hole phenomenon and changes in the earth's upper atmosphere

A scientist of Calcutta University studied the radio noise in HF and VHF bands which affect communication between India and Antarctica. He also studied the integrated water vapour content of the atmosphere around Maitri station which severely affect radio communication. The millimetre wave satellite and ground communication systems have been proposed for use in Antarctica.

Medical Sciences

Continuing its research programme in Antarctica, AllMS carried out. Studies on the physiological changes and their correlation with circadian rhythm alterations in Antarctic environment, effect of behavioural modulators on the change: in body system in Antarctica and on the effect of cold exposure (cold adaptation) in Antarctica.

Meteorological Studies

IMD scientists in active collaboration with meteorologists of IAF and Indian Navy, during the austral summer, carried out routine three hourly synoptic weather observations, sunphotometer observations, and took ozonesonde and radiosonde ascents.

Communication

As in previous years the bulk of the responsibility for communication between Antarctica and India rested on the shoulders of communication team of Indian Navy. HF and VHF communication links were established between ship-Maitri station-Dakshin Gangotri-Payer camp and the flying helicopters. HF voice and data communications were tried on Amateur bands with encouraging results, Communication officer also prepared an inventory of the equipments lying in Maitri/DG area.

R & DE (E) engineers in collaboration with DEAL scientist successfully established voice and data link over HF with India using different modes. A patch up of HF signals from Antarctica with the telephone network in India on an experimental basis also proved to be successful. This opens a new era of communication between Antarctica and India at practically no cost.

Life support Systems

Engineers of R&DE(E) and Indian Army tarried out studies on various life support systems within and outside Maitri station. A new power house, was made operational, the infrastructure of which was created during the previous expedition. The plant housed in a 20m x 8m x 4.5m prefabricated accommodation, contained 6 nos. of 62.5 KVA and 2 nos. of 125 KVA diesel gensets.

While trying to improve the central heating system at Mattri station, R&DE(E) engineers successfully reduced the corrosion tendency of pipelines by corrosion preventive treatment without using Mono Ethylene Glycol (MEG).

Two fire barrier walls were fixed inside the station, one between the boiler room and kitchen and another between genset room and workshop in A-Block, Further, a fire cheek door was installed in the Mam Block of the building.

Columnar support was provided to the portion of the water supply pipeline filiating in Priyadarshini Lake at the pimiphouse end. Eight numbers of columns were erected resting on lake bed. Each of these has 1.5 m dia size load bearing area at the base. These columns are supporting the pipeline over a length of 12 m which was earlier floating on sealed empty fuel barrels. This portion of the superstructure was further strengthened by providing cross channel members. Surface floor area of superstructure has also been increased for ease of pipeline maintenance.

Other Logistics

Routine maintenance of station was carried out while the previous wintering team was still around. The work done included cleaning of both the klargesters, flushing of pipelines of central heating system and changing of the coil in one of the Boilers. Vehicles which were badly in need of repairs, were also attended to, making it possible to undertake the joint convoy of Eighth and Ninth wintering teams.

R&DE(E) engineers repaired the refrigeration system of Maitri station that had gone out of order.

For the summer camp at Maitri, additional facilities have been created. These include construction of three living modules, two bath modules and one toilet module.

Erection of a 28 m high mast for scientific studies and communication purposes has been a highlight of the expedition. This goes side by side with the erection of Log Periodic Antenna (LPA) and inverted vee as well as dipole antennae for HF communication.

The members of the Army contingent, besides providing all the logistics mentioned above, also handled the cargo throughout the duration of the expedition.

Catering services were rendered by the Indian Navy during the expedition. Till the time ship left for homebound journey, four cooks provided food for the entire team on board ship and at Maitri. Naval meteorologist of the contingent acted as the mess secretary on board ship while at Maitri a scientist from SOI was entrusted with the same responsibility.

Helo Operations

During the eleventh expedition several records in flying of the helicopters were set despite less number of flying days available due to bad weather conditions. MI-8 helicopters of IAF flew for a total of 168 hours making 328 sorties. They carried a total of 483.05 tonnes of load. MI-8 flew to a record distance of 216 kms south of Maitri in a single sortie upto 72°36'30"S latitude on 23 February 1992 for meteorite search, recce of Conrad mountains and a visit to the memorial site in Humboldt mountains.

Chetak helicopters of Indian Navy flew for a total of 339 hours 25 minutes which is a record as far as their flying in Antarctica is concerned. It is also, probably for the first time, that Chetak helicopters have flown in all the mountains within range from Maitri in a single expedition. These are Payer, Weyprecht, Gruber, Petermann I,II,III and Humboldt. Chetaks also created a record when on 02 February 1992 they clocked 19 hours of flying in a single day carrying 4.5 tonnes of cargo.

The Ninth Wintering

With the hazy Sun feeling shy behind a screen of fog gently stretched over the ice shelf, MV Thuleland turned homeward at 1700 GMT on 25 February 1992, with 25 members of VIII winter team and 72 members of XI expedition summer team on board. Thus began the-ninth wintering-by Indians in Antarctica and the fourth in Maitri. Thereafter, the team faced unforeseen circumstances which tested to its limits, the endurance, will and determination of the members for survival

Winter Tasks

The winter tasks can be divided into two - Scientific and Logistic.

Scientific

1. Geolpgy/Glaciology : Geological field work in Gruber massif of Wohlthat mountain chain, fixing of stakes in polar ice for ice dynamics studies and drilling for ice core to collect data on palaeoclimate were the aims set for the winter team. However, burning of some vital parts of Geo Tech Ice Core Drill Machine in a fire incident on 03 April 1992 did not allow us to undertake the drilling work till following summer when the lost parts were replenished. An initial effort to drill in the polar ice with this machine met with partial success. The outcome, however, prepared us for the problems to be encountered when drilling was to be taken up full fledgedly during next winter.

Eleven stakes were fixed in December 1992 along the usual Maitri - Dakshin Gangotri convoy route, in Baalsrudfjellet - Sonstebynuten sector, south of Maitri. They were fixed at intervals of approximately 1.5 km and their coordinates noted with the help of MX 4102 Satellite Navigation system.

Snow level over Dakshin Gangotri ice shelf was also recorded twice, in March and November 1992, which showed net accumulation during the period.

Due to a change in priority of tasks coupled with some medical reasons, no attempt was made to reach the distant mountains viz. Gruber massif. Instead, Schirmacher Oasis and three of the nunataks exposed south of it were studied for occurrence of alkali rocks. Approximately 150 samples were collected for further studies. In addition, a total of 27 nos. of bulk sediment samples including moraine and lake sediments from these areas were collected for qualitative and quantitative analyses to determine presence of heavy metals, if any.

2. Meteorology : The meteorological observational programme was mostly on-going in nature. Besides the regular three hourly synoptic observations and their real time transmission to Global Telecommunication Network via New Delhi, at least two cloud imageries were received daily from the polar orbiting NOAA satellites.

A total of 45 ozonesonde and 19 radiometer sonde ascents were taken during austral winter. Two new programmes - Low Level Slow Rising Radiosonde (LLSR-RS) ascents and Pilot Balloon (PB) ascents were taken up during the expedition. In all 12 LLSR and 5 PB ascents were taken in different months.

3. Geomagnetism: Continuous observation and recording of geomagnetic field changes and maintenance of the geomagnetic observatory at Maitri went on side by side during winter. In addition, a successful attempt was made to study the HF communication conditions and auroral activity vis-a-vis magnetic variations.

4. Atmospheric science: In the absence of any wintering member from NPL, the responsibility of collecting data to probe the Planetary Boundary Layer upto an altitude of 1 km, collecting air samples to study atmospheric pollution and data on various parameters for studying ABL characteristics, was shouldered by IIG scientist.

5. Experimental HF data link : The HF data link established between Maitri and India during austral summer was maintained during austral winter, Data communication was successfully carried out using FEC and AMTOR modes. Trials were also made in PACKET mode.

6. Experiments on Amateur Radio Bands: Though no such task was assigned, the communication officer of the team carried out experiments on amateur radio bands using his personal HAM radio set. The success achieved has been very encouraging. ARQ, FEC_f ASCII, BAUDOT and fax pictures over HF were tried out. Also, for the first time, teletype link on FEC was established successfully with the ship after its departure from Goa with the XII expedition team.

7. Computer work : Stuck inventory of entire Maitri station has been updated and converted into computer database. This has helped in making a comprehensive account of the material available at Maitri. A rule base for troubleshooting the boilers was developed to upgrade the already existing 'Expert. System'. Information and data on other systems such as plan drawings of Maitri station, boiler room layout, etc. were created using AUTOCAD which would prove to be guiding factors for future maintenance and expansion of these systems. Various computer programmes were written to record the data collected by IMD and IIG scientists.

8. Medical research : The data for research projects of AIIMS were collected by the wintering doctor from Army Medical Corps. This involved a) collection of four hourly blood samples of seven members at four monthly intervals, b) project on effect of bright light exposure during polar nights, c) project on effect of magnetic storm on blood and urine biochemistry, d) two monthly psychological assessment of all members, e) special psychological assessment of five members done quarterly and f) special monthly health status evaluation and ECG of all members.

Logistic

Logistic support to the winter team was provided by personnel from Indian Army and Indian Navy, Army contingent carried out maintenance of life support systems at Maitri, carried fuel and other stores from Dakshin Gangotri/shelf edge to Maitri by surface transport and provided various types of

logistic support to the scientist members helping them to carry out their tasks. The medical care was also extended by a specialist surgeon from DGAFMS.

Naval team looked after the maintenance of all communication systems and ensured a continuous link with India and the rest of the world. Besides, the important task of cooking was taken care of by a Naval cook. Navai team also provided expertise in the maintenance of and repairing vital electrical and electronic equipments/gadgets in the station. They also provided communication facilities to field parties and convoys.

Some of the highlights of logistic work done during the ninth wintering are:

- a) thorough cleaning of the station loft giving it an aesthetic look while completing the task of fixing gypsum boards as fire safety measure,
- b) providing separate video room and library with proper sitting arrangements in independent bays of the loft,
- c) converting one bay in the loft into a decent guest room,
- d) providing wooden shelves in the general stores and systematically arranging and making inventory of the items in stock,
- e) painting the walls and roof of station common lounge,
- f) making a fresh and systematic catalogue of all the books in the library,
- g) thorough cleaning of the campus area around Maitri.
- h) providing water hoses with suitable connections at strategic points inside the station building to fight any eventuality caused by fire,
- i) extensive repair work on the previously grounded snow mobiles which resulted in handing over of all surface transport vehicles like PB-170s, PB - 270s, Snocats etc. in good working condition to the tenth wintering team,
- j) electrical rewiring of major portions of Maitri station.

Part of the major repairs of vehicles included replacement of engines in two SNOCATs and one PISTEN BULLY and putting on road the APC Topas vehicle with improved and improvised wind shield fitted with a snow wiper for better viewing and convenient driving. The Topas was also provided with heating system for crew cabin. This vehicle can now be used for field work over rocky terrains and also for rescue operations.

The ramp of the hangar at Dakshin Gangotri was covered by a column of six metres of solid ice. This ice column was removed in November 1992 to extricate two vehicles - a PB 170 and one snow cutler besides a snow scooter. These were retrieved and repaired for use although the snow culler needs further repairs.

Maitri station was handed over to the new team on 01 February 1993 in an excellent condition, Dr.V.K.Dhargalkar and his tenth wintering team took charge of the operations after a month long briefing of their logistic personnel and joint handling of life support systems at the station for sometime by members of both the teams.

Good Bye Antarctica

The ninth winter team along with members of the XII expedition summer component left 'India Bay' on board MV Thuleland on 18 February 1993. The ship reached Atka Bay in West Antarctica on 22 February morning to pick up four Germans and some containers from their station G.Von Neumayer. We sailed homeward the same night and were promptly caught up in a cyclone for three days. German expedition members disembarked and their containers were off loaded at Port Louis, Mauritius on 10 March 1993. During the return trip, scientific studies were carried out in the form of oceanographic research, iceberg monitoring in cold waters and meteorological observations.

"Between the idea and the reality, Between the motion and the act, Falls the shadow—" yet, the team that arrived at Goa on 22 March 1993 was a contented lot satisfied with the efforts it put in in that land of perpetual challenge.

References

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- Mukerji,S.(1993):Unpublished Report of the Work Done During Ninth Wintering at Maitri, Schirmacher Oasis, Antarctica. Department of Ocean Development, New Delhi.



Fig 1: Fixing VHF antenna at the tap of 28 m tower—July 1992.



Fig 2: Fixing fire barrier between boiler mom and Kitchen block — May 1992.



*Fig. 3: Volunteering For Cold Stress test --- January 1992.
(Photo Coutrtesy: Bimal Chhajjer)*



Fig 4: Releasing Meterological Balloon After A Snow Fall ---- March 1992.



Fig 5 : water Supply Pipeline Drawn From Lake Priyadarshini.



Fig 6 : Replacing a new engine in PB- 270 Snow mobile ---- June 1992.



Fig 7: Skua With Chicks in Schirmacher Oasis --- November 1992.



Fig 8: A Skua nest with remains of preyed birds in Schirmacher Oasis -- November 1992.



*Fig 9 : Extending The Entry Shaft of Burried Hangar in Dakshin Gangotri
_ November 1992.*



*Fig 10: Removing Thick Pile Of Snowlice to open D.G. hangar and retrieve two - Vehicles
from within --- November 1992.*



Fig 11: Front Side View of Maitri Camp --- May 1992.



Fig 12: Clearing Snow accumulation from klargest area on the western said of Maitri Station



Fig 13: Clearing Snow Accumulation Outside Generator block, on the Southern Side of Maitri station.



Fig 14: Basic dykes forming small swarm in Baalsrudfjellet nunatak south of Schirmacher Oasis.



Fig 15: Snout of Dakshin Gangotri Glacier.



Fig 16: Green house in full bloom — December 1992.